11/13/2025

COMPUTER NETWORK

LAB 13

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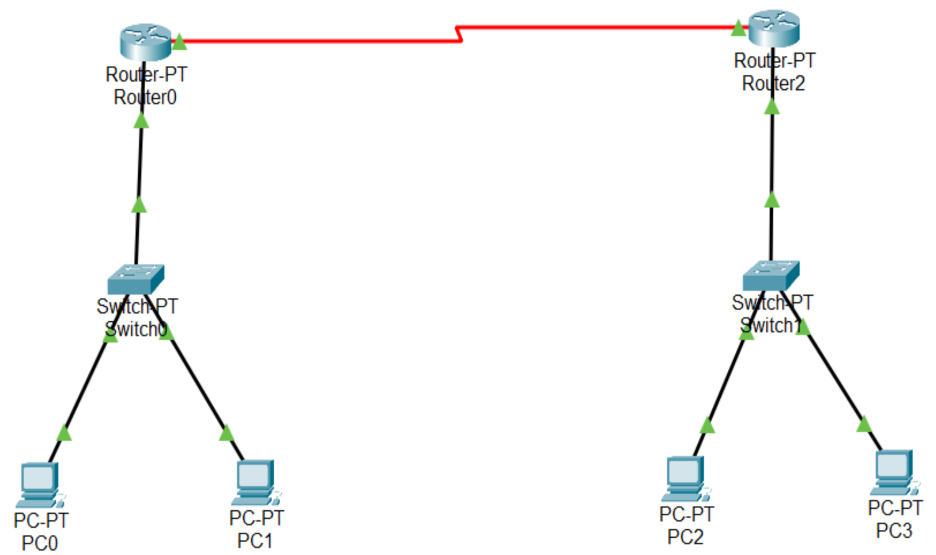
SAP 64281

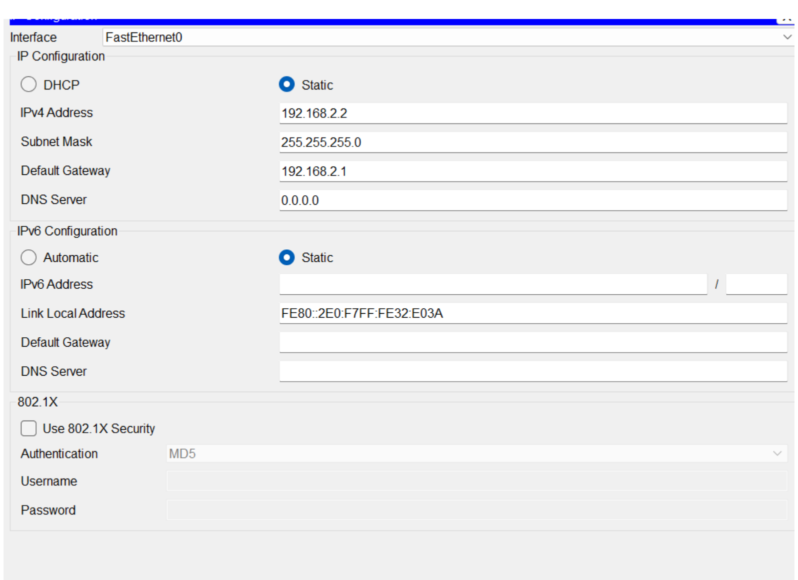
TOPOLOGY CREATED:



HOW:

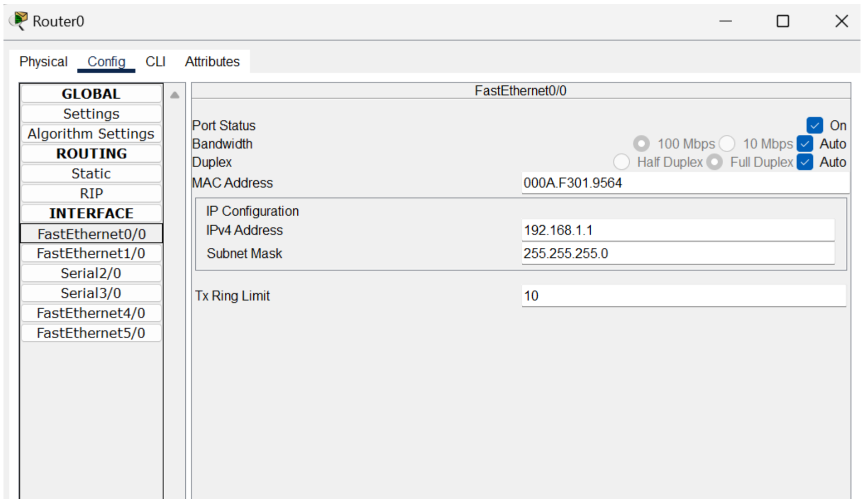
Step 1:

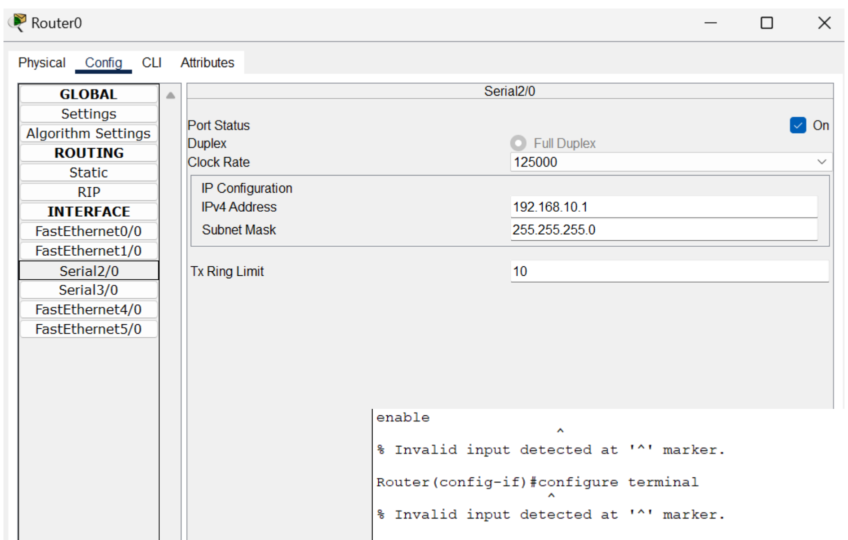
Configure all PCs as below 1.



Step 2:

Configure router0:

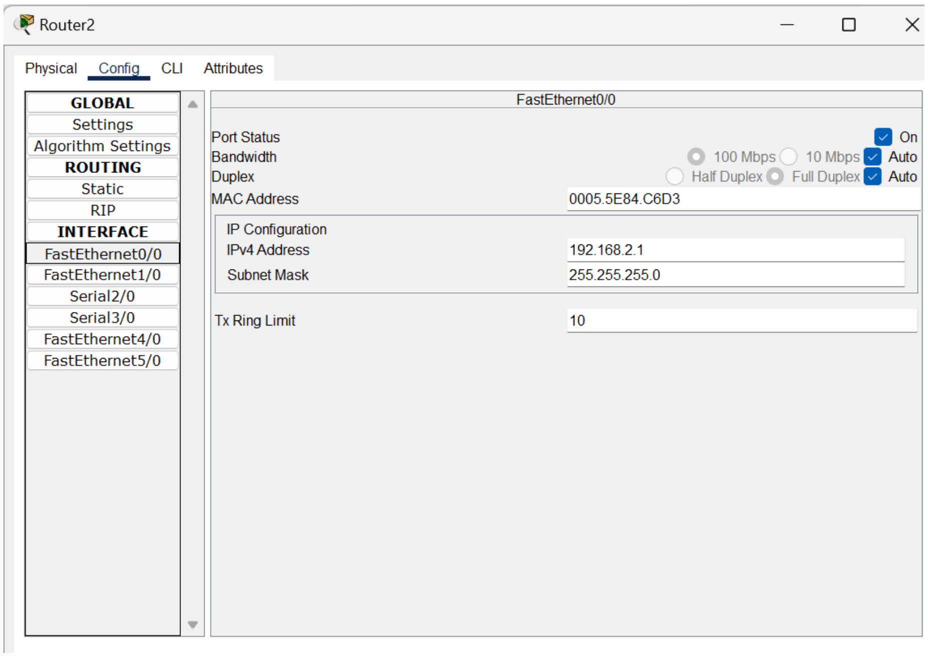


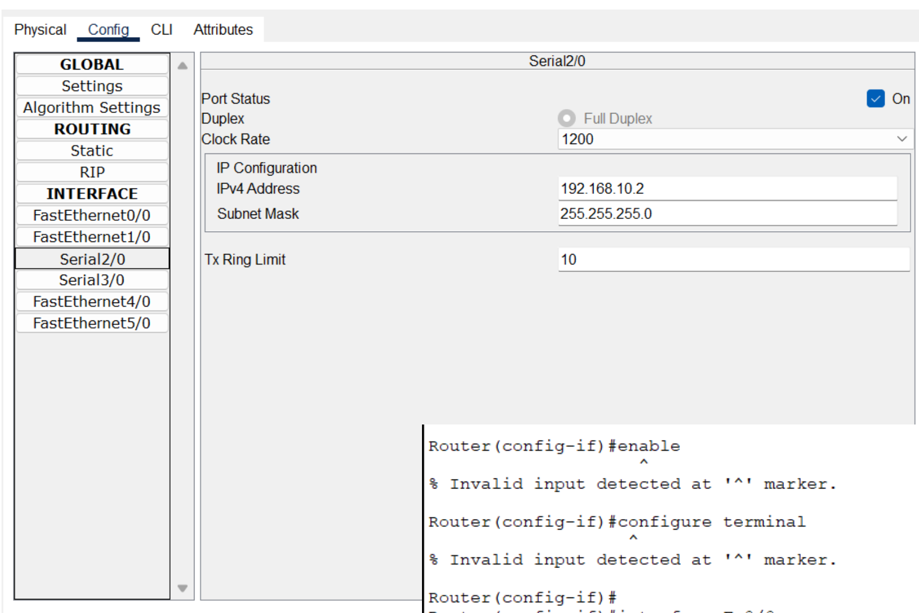


Configuration of CLI of router0:

```
enable
                     ^
% Invalid input detected at '^' marker.

Router(config-if)#configure terminal
                    ^
% Invalid input detected at '^' marker.

Router(config-if)#
Router(config-if)#interface Fa0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#
Router(config-if)#interface Se2/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#! If this serial interface is connected to the DTE on the other router,
Router(config-if)#! Packet Tracer will show one end as DCE  set clock rate on the DCE side:
Router(config-if)#clock rate 64000      ! run this command while still in interface Se2/0 if
this router is DCE
                                        ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#
Router(config)#! Add a static route for the remote LAN:
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.10.2
Router(config)#
Router(config)#end
Router#writememory
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```

Configure router1:

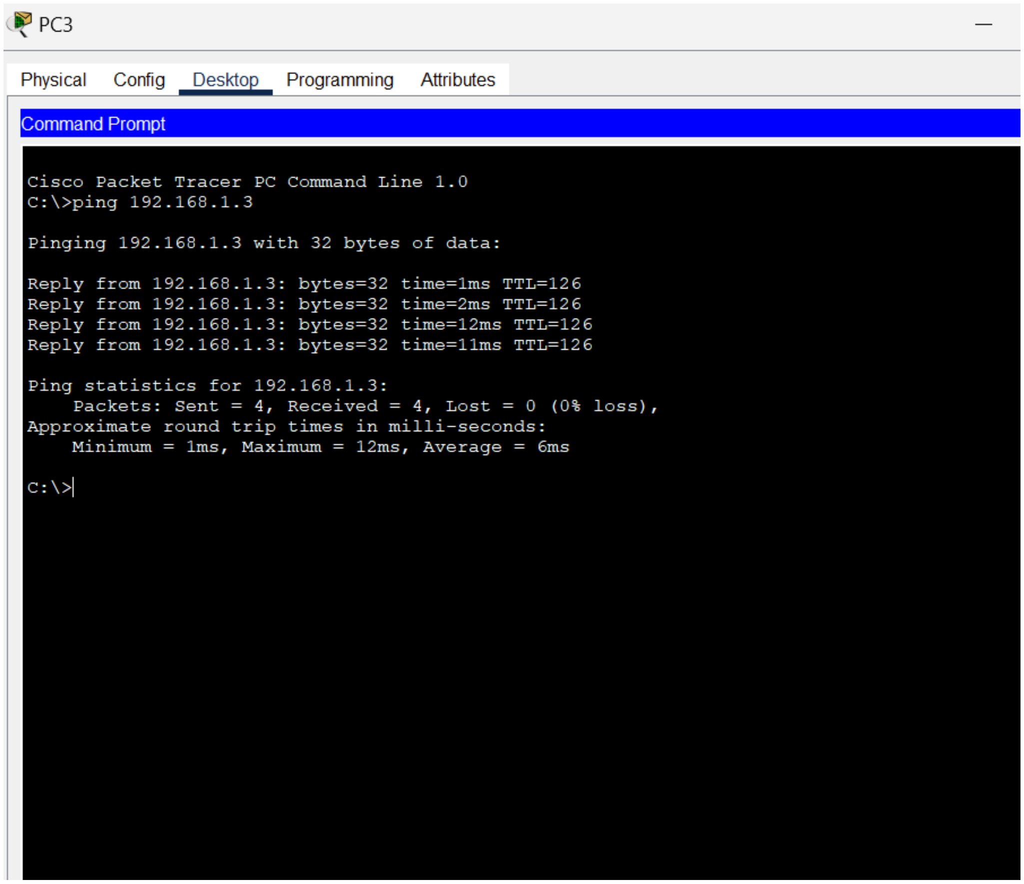




Configuration of CLI of router1:

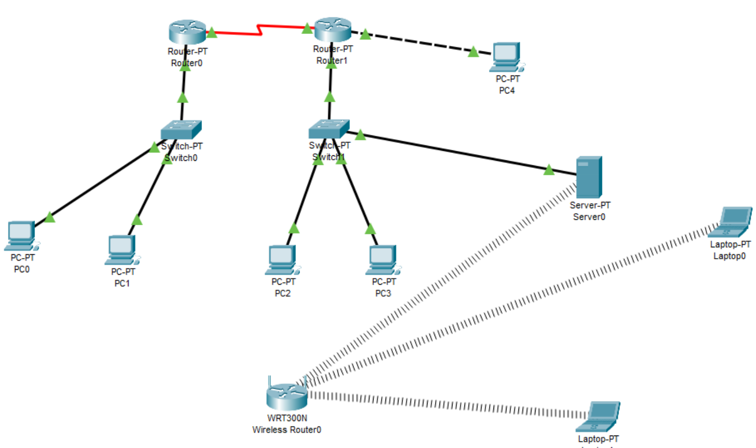
```
Router(config-if)#enable
                    ^
% Invalid input detected at '^' marker.

Router(config-if)#configure terminal
                   ^
% Invalid input detected at '^' marker.

Router(config-if)#
Router(config-if)#interface Fa0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#
Router(config-if)#interface Se2/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#! Only set clock rate here if this end is the DCE cable end:
Router(config-if)#clock rate 64000    ! (only on the DCE end)
                                      ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#
Router(config)#! Static route to left LAN:
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)#
Router(config)#end
Router#writememory
%SYS-5-CONFIG_I: Configured from console by console
```

Ping from pc3 to pc1:

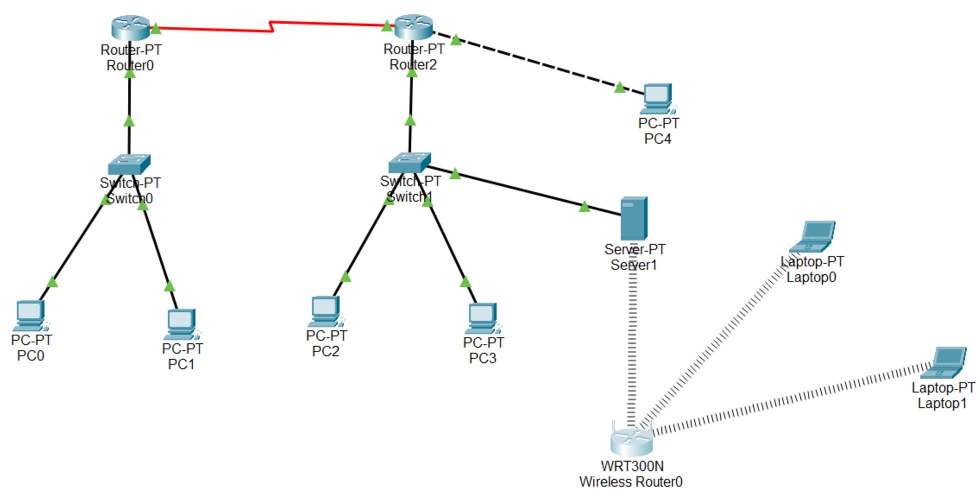


TASK 2:

Create the topology below using default routing. Take screenshot of each step and show the successful ping of the packets from one pc to another pc.



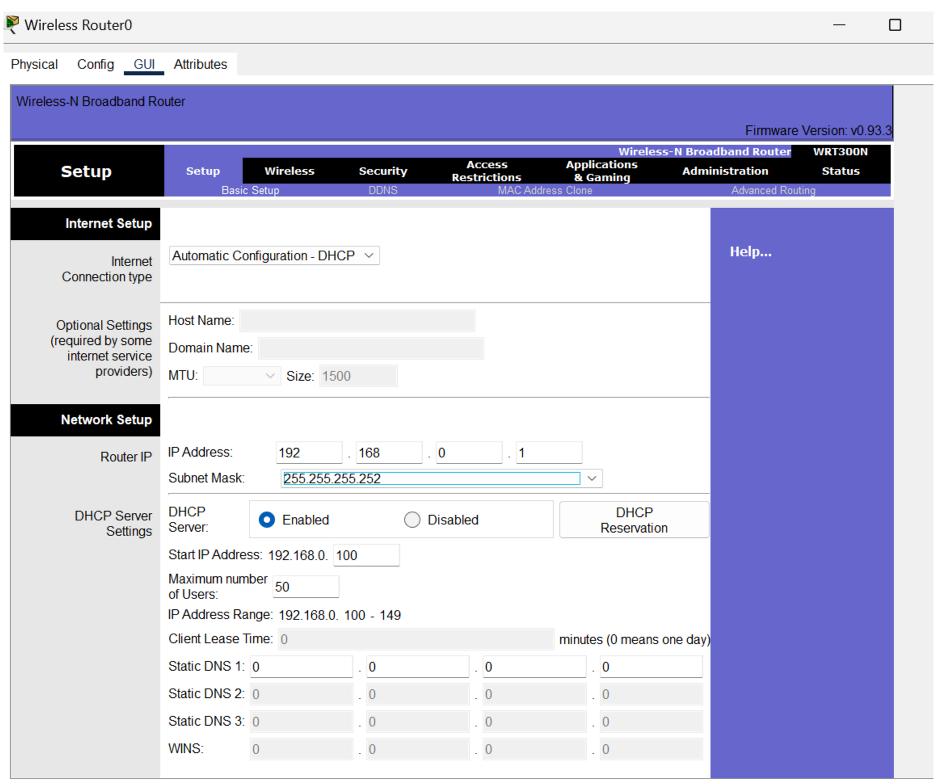
Topology created:

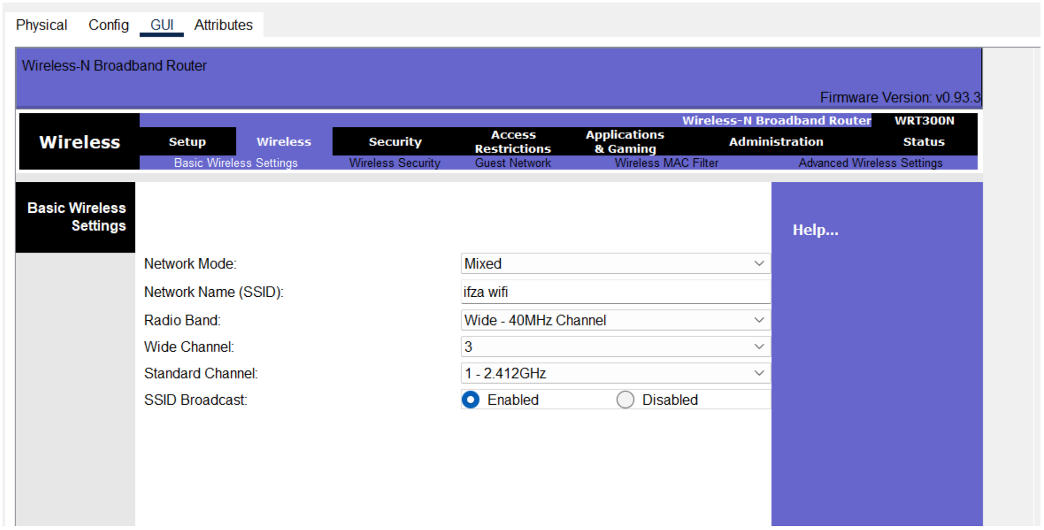


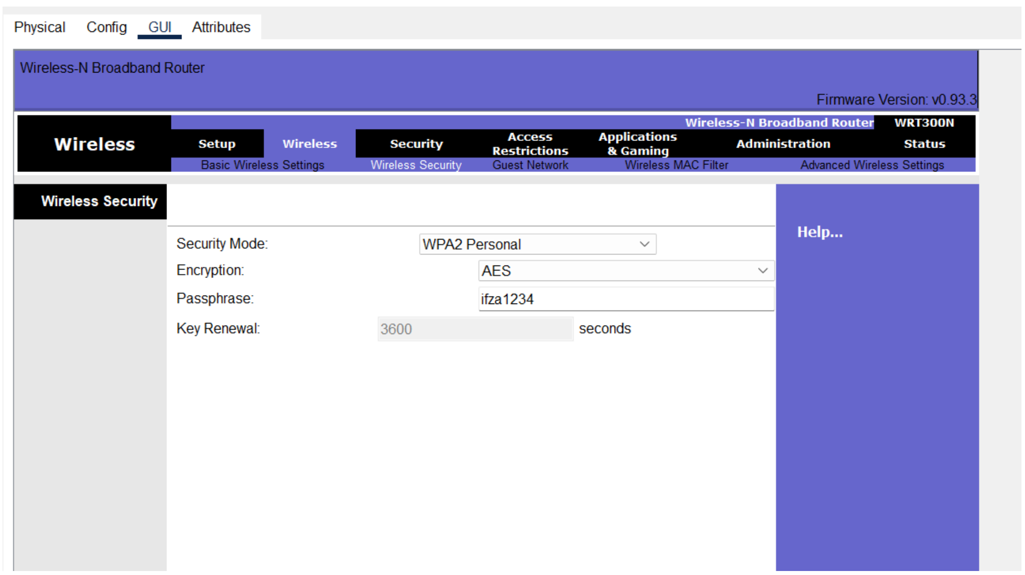
Half steps are same as above some steps are added:

Step1:

Configure a wireless router.



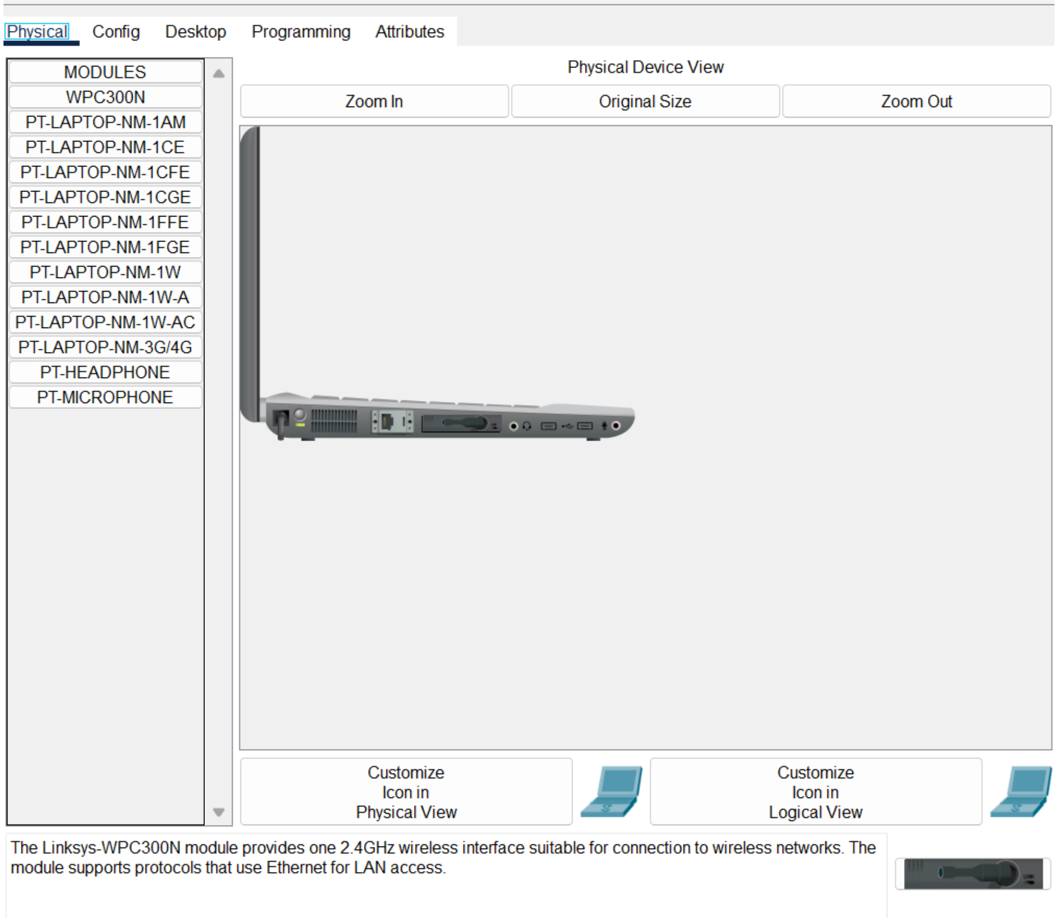




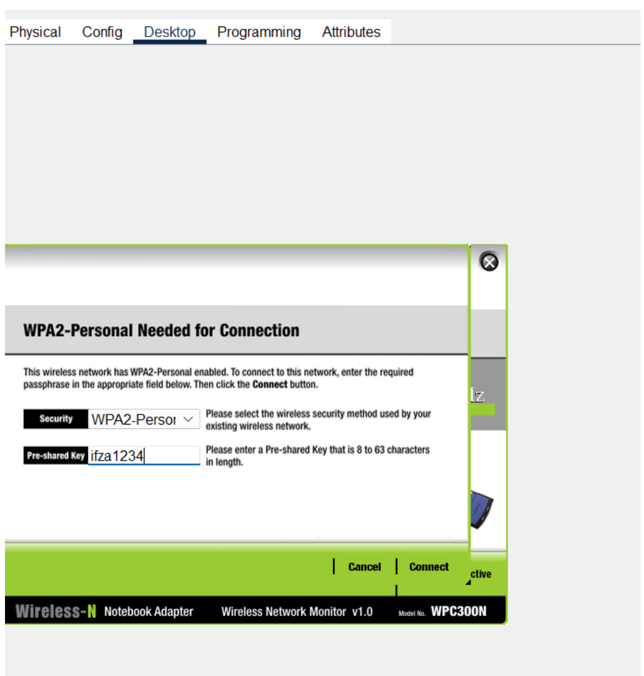
Step2:

Now set laptop for wireless connection.

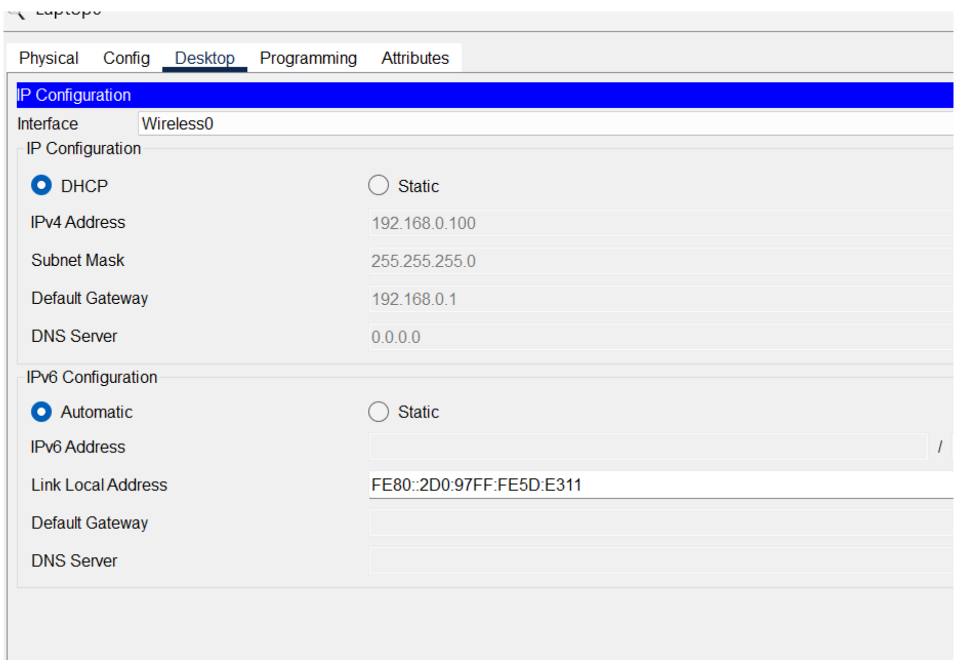
From below pic of physical tab off button of laptop remove vaccum and add WPC300N vaccum and On the laptop.



Now move to desktop->wireless find network of wireless router I save as ifzawifi shown in tab then try to connect it ask for password add the password that is in the router as in below.

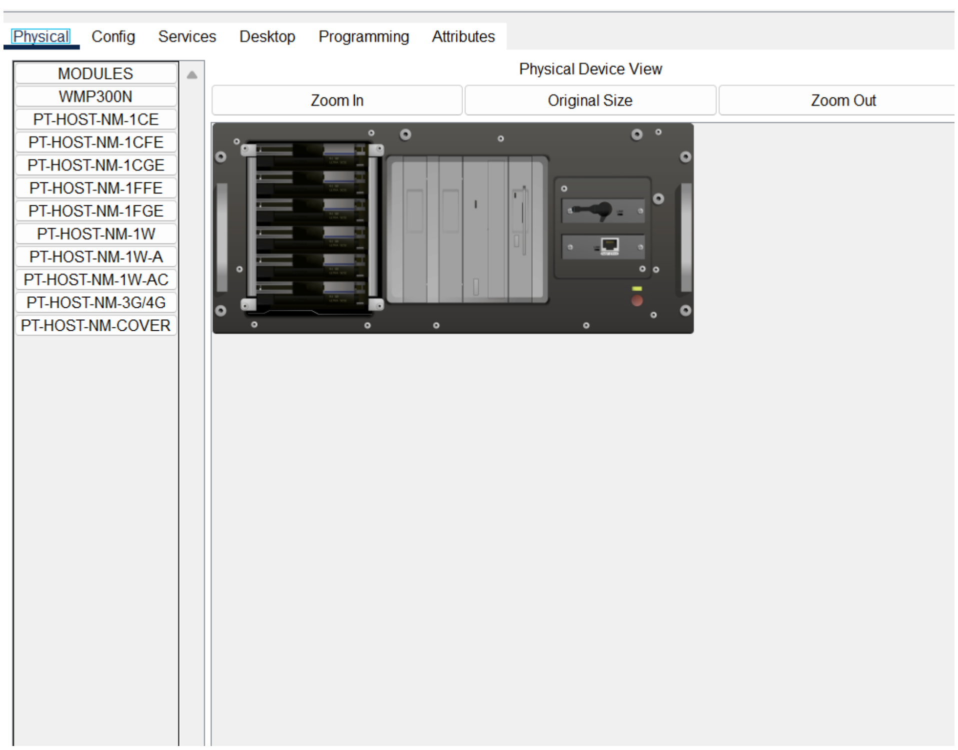


Now get the DHCP ip wirelessly through the router as in below.

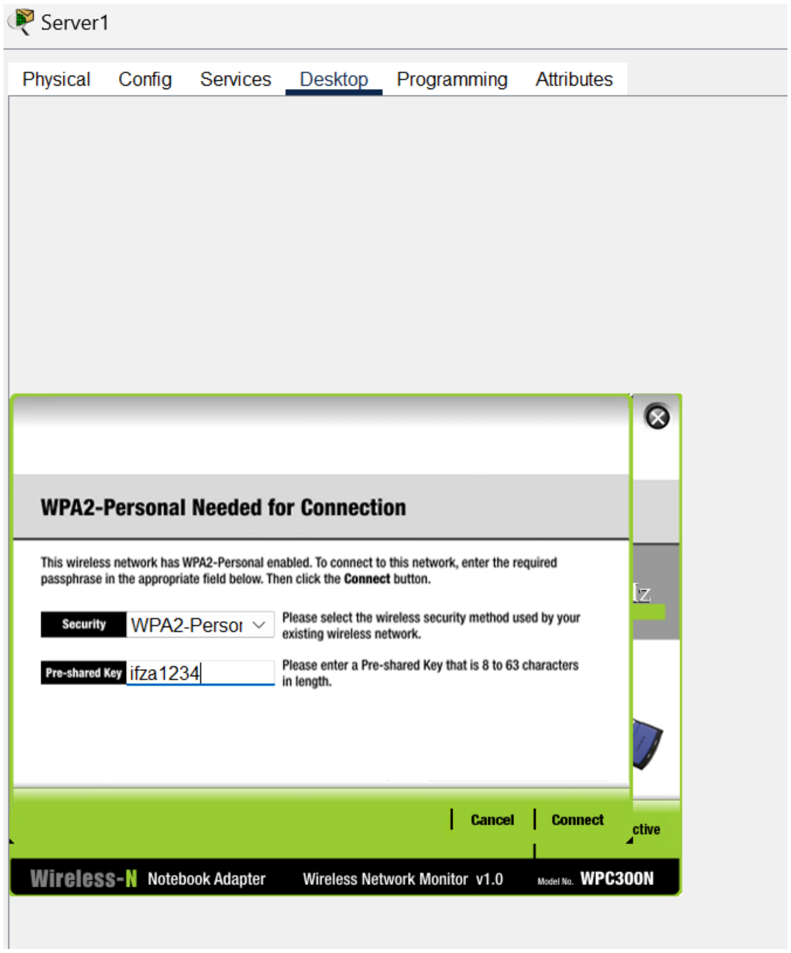


Step 3:

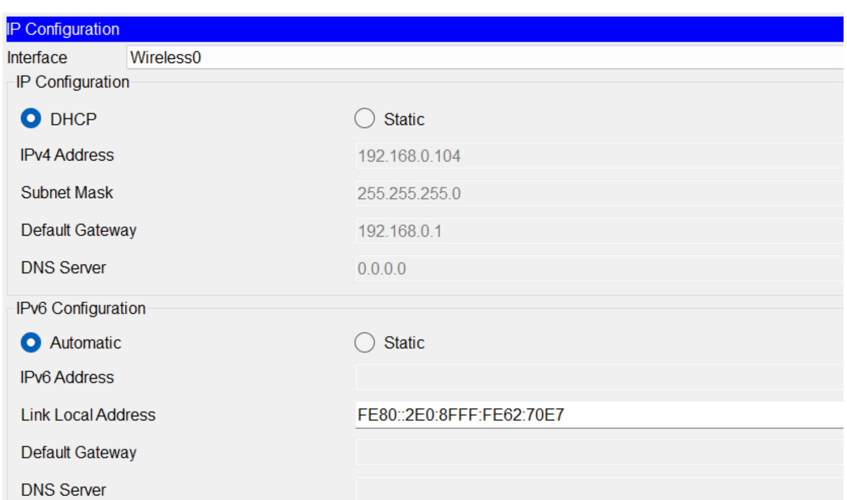
Now configure server so that it can connect in both physically and wirelessly through the wieless router and pt router at the same time. Add 1CFE and WMP300N both in places after turn off the server then again turn on the server.



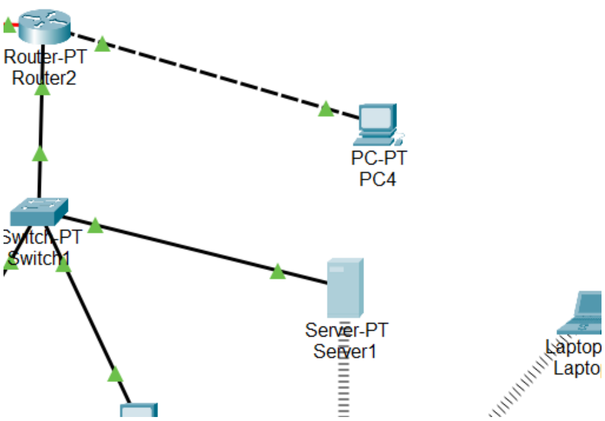
now connect the server wirelessly from desktop->wireless same as the laptops.



Attain DHCP ip of server via wireless router.

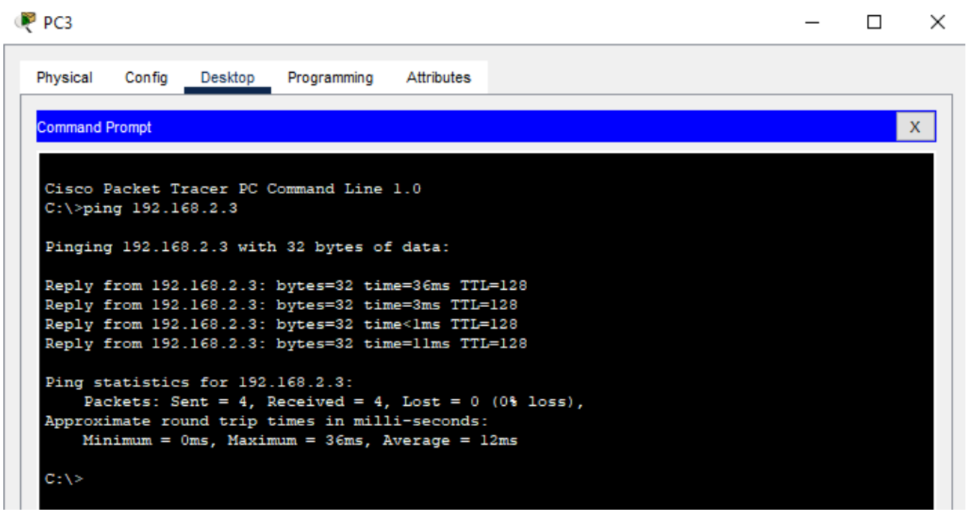


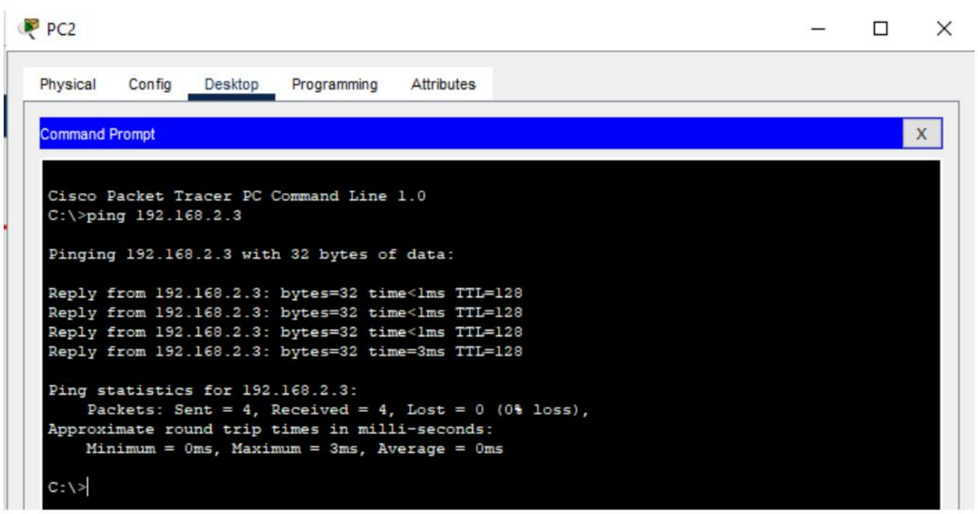
Add an additional pc to pt router 2 directly not via switch.



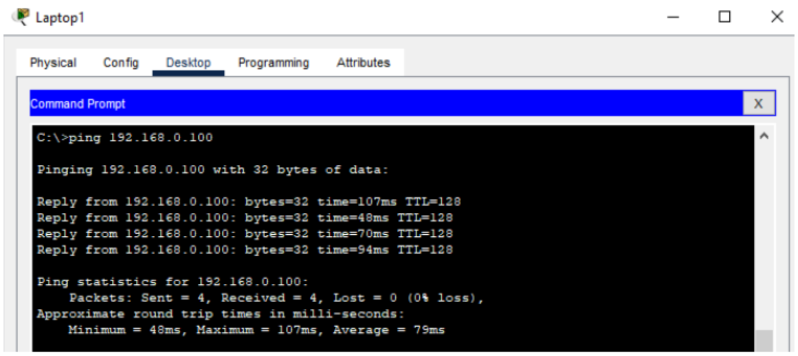
Last step:

Now ping the network for checking the valid working of the network.

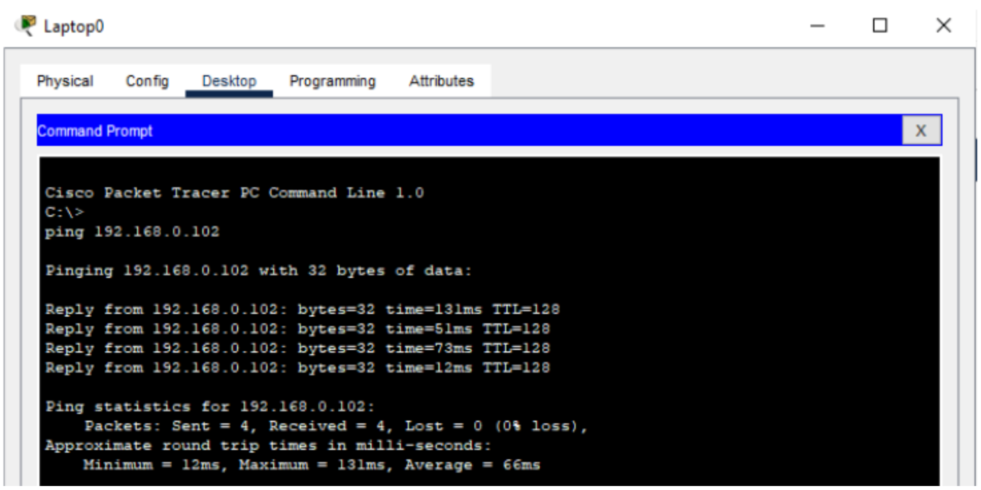




Pinging laptop0 from laptop1:



Pinging laptop1 from laptop0



THE END